Criteria for Diagnosis and Staging of Alzheimer's Disease

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Alzheimer’s Association Business Consortium (AABC)

Global Biomarker Standardization Consortium (GBSC)

Global Alzheimer’s Association Interactive Network

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Alzheimer’s Disease Neuroimaging Initiative Private Partner Scientific Board (ADNI-PPSB)

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Appropriate Use Criteria for Amyloid and Tau PET Imaging
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Starting in 1984, the Association has been at the forefront of the criteria to diagnose Alzheimer's disease. An Alzheimer's Association workgroup and National Institutes of Health workgroup (NINCDS-ADRDA) was the leading criteria for trials and diagnosis in the clinical setting, until the 2011 National Institute on Aging and Alzheimer's Association (NIA-AA) clinical guidance. Building off this, the 2018 NIA-AA research framework provided context of current scientific knowledge to outline a framing for research needs and hypotheses to be examined closer.

"Revised Criteria for Diagnosis and Staging of Alzheimer's Disease" was published June 27, 2024 in *Alzheimer's & Dementia®: The Journal of the Alzheimer's Association*, with corresponding commentaries published in Nature Medicine and Nature Aging. The 2024 version incorporates the latest in science to define more fully the criteria for diagnosis and staging of Alzheimer's.

Diagnosing Alzheimer's disease is advancing with science. By incorporating new scientific insights and technological advances, these new criteria aim to:

- Improve current diagnosis, including accuracy.
- Provide context for a biological definition that will inform the next generation of clinical trials.

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Overview

The 2024 criteria provides an outline for the diagnosis and staging of Alzheimer's disease. Several core principles emerged from these efforts, including:

- Alzheimer's disease should be defined biologically, not based on a clinical syndrome(s).
- The disease is a continuum that begins with the appearance of changes in the brain associated with the disease processes in asymptomatic individuals, which progresses through stages of increasing levels of disease-related brain changes, eventually leading to the appearance and progression of clinical symptoms.
- The disease is diagnosed in individuals by abnormalities on core biomarkers.

The criteria do not support the evaluation of Alzheimer's disease-related brain changes in asymptomatic individuals for purposes of clinical care at this time — i.e. outside the context of observational or therapeutic research studies.

In early 2022, the Alzheimer's Association convened a steering

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Under the leadership of Dr. Jack, the workgroup presented their work at several scientific conferences, including the Alzheimer's Association International Conference® (AAIC®) 2023, at Clinical Trials in Alzheimer's Disease (CTAD) 2023 and at AD/PD 2024.

Why update the criteria now?

In a new era where treatments that address the underlying biology of Alzheimer's disease are available and will continue to emerge, the present document has progressed from a framework for research to criteria for diagnosis and staging that are intended for clinical use as well as research.

Validated biomarkers in the 2018 framework were based on either cerebrospinal fluid (CSF) assays or imaging. Since then, some plasma-based biomarkers with excellent diagnostic performance have been developed and are being clinically validated. The 2024 criteria have correspondingly incorporated plasma biomarkers into updated criteria for biomarker categorization, disease diagnosis and staging.

Lastly, research studies have demonstrated that imaging and fluid biomarkers within a category — although often concordant — are not interchangeable in many clinical scenarios. In the present document, the workgroup has updated biomarker classification criteria to accommodate nonequivalence between fluid and imaging biomarkers within a category.

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begin.

According to the criteria, “[T]he clinical use of [Alzheimer’s] biomarkers is presently intended for the evaluation of symptomatic individuals, not cognitively unimpaired individuals. ... At the present time disease-targeted therapies have not been approved for cognitively unimpaired individuals with [Alzheimer’s]. For this reason, we currently recommend against diagnostic testing in cognitively unimpaired individuals outside the context of ... research studies.”

The Association is initiating a collaboration with clinical and subject-matter experts, methodologists, external organizations, and patient representatives to develop guidelines around the clinical implementation of Alzheimer’s disease staging criteria and treatment. This work is beginning later in 2024.

2022-2024 workgroup members: Revised criteria for diagnosis and staging of Alzheimer's disease

Workgroup members were selected to achieve a range of scientific expertise, the broad representation of different institutions (public, academic and private) and professional organizations involved with Alzheimer's research, and geographic and gender diversity. Given the importance of the criteria to support research studies testing clinical interventions, scientific expertise

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- Maria C. Carrillo, Ph.D., Alzheimer's Association, convener and steering committee
- Billy Dunn, M.D., Independent, steering committee
- Ana Graf, M.D., Novartis
- Oskar Hansson, M.D., Ph.D., Lund University
- Carole Ho, M.D., Denali Therapeutics
- Clifford R. Jack Jr., M.D., Mayo Clinic, chair and steering committee
- William Jagust, M.D., University of California, Berkeley
- Eliezer Masliah*, M.D., National Institutes of Health, steering committee
- Eric McDade, D.O., Washington University in St. Louis
- José Luis Molinuevo, M.D., Ph.D., Lundbeck
- Ozioma Okonkwo, Ph.D., University of Wisconsin, Madison
- Luca Pani, M.D., University of Miami, Former Italian Regulatory Agency
- Michael Rafii, M.D., Ph.D., University of Southern California
- Laurie Ryan*, Ph.D., National Institute on Aging
- Phillip Scheltens, M.D., Ph.D., Life Science Partners

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View each workgroup member's disclosures (PDF).

Previous criteria, guidance and frameworks

Each workgroup initially issued proposed recommendations that were posted for public comment. The final versions of the guidance, revised to reflect input from the professional community at large, appear as free-access papers in *Alzheimer's & Dementia®: The Journal of the Alzheimer's Association*.

National Institute on Aging (NIA) — Alzheimer's Association Diagnostic Guidelines Focusing on the Three Stages of Alzheimer's Disease (2011)

By incorporating new scientific insights and technological advances, the 2011 guidance aimed to improve current diagnosis, strengthen autopsy reporting of Alzheimer's brain changes and establish a research agenda for future progress in earlier detection and even greater diagnostic accuracy.

The workgroups developed guidance that focused on the three stages of Alzheimer's disease:

1. Preclinical (asymptomatic or asymptomatic) Alzheimer's

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Clifford R. Jack Jr. et al. "Introduction to the recommendations from the National Institute on Aging – Alzheimer's Association workgroups on diagnostic guidelines for Alzheimer's disease." *Alzheimer's & Dementia: The Journal of the Alzheimer's Association* 2011;7(3):257 – 262.

Preclinical (asymptomatic) Alzheimer's disease

This is a newly defined stage of the disease reflecting current evidence that measurable biomarker changes in the brain may occur years before symptoms affecting memory, thinking or behavior can be detected by affected individuals or their physicians. While the proposed guidance for preclinical Alzheimer's disease identifies these preclinical changes as an Alzheimer's stage, they do not establish diagnostic criteria that doctors can use now. Rather, they propose additional research to establish which biomarkers may best confirm that Alzheimer's-related changes are underway and how best to measure them.

Reisa A. Sperling et al. "Toward defining the preclinical stages of Alzheimer's disease: Recommendations from the National Institute on Aging – Alzheimer's Association workgroups on diagnostic guidelines for Alzheimer's disease." *Alzheimer's & Dementia: The Journal of the Alzheimer's Association* 2011;7(3):280 – 292.

Mild cognitive impairment (MCI) due to Alzheimer's disease

In this stage, mild changes in memory and thinking are noticeable and

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Marilyn S. Albert et al. "The diagnosis of mild cognitive impairment due to Alzheimer's disease: Recommendations from the National Institute on Aging – Alzheimer's Association workgroups on diagnostic guidelines for Alzheimer's disease." *Alzheimer's & Dementia: The Journal of the Alzheimer's Association* 2011;7(3):270 – 279.

Dementia due to Alzheimer's disease

In this stage, impairments in memory, thinking and behavior decrease a person's ability to function independently in everyday life. This guidance for dementia due to Alzheimer's disease updates and clarifies clinical criteria to diagnose dementia from all causes and specifically from Alzheimer's disease. These criteria are sufficiently broad and flexible to be used now both by community practitioners without access to neuropsychological testing, specialized brain imaging, or cerebrospinal fluid testing and by specialists engaged in research or clinical studies who have access to such tools.

Guy M. McKhann et al. "The diagnosis of dementia due to Alzheimer's disease: Recommendations from the National Institute on Aging – Alzheimer's Association workgroups on diagnostic guidelines for Alzheimer's disease." *Alzheimer's & Dementia: The Journal of the Alzheimer's Association* 2011;7(3):263 – 269.

National Institute on Aging (NIA) – Alzheimer's Association Guideline on Neuropathologic Assessment of Alzheimer's During an Autopsy

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Ranking the severity of Alzheimer's pathology based on three hallmark changes.

- Reporting these rankings as "Alzheimer's disease neuropathologic changes," whether or not the person was ever diagnosed with Alzheimer's during life, with a goal of understanding the full range of brain changes that may occur in people with or without Alzheimer's symptoms.

- Including assessment of Lewy bodies, vascular abnormalities and other brain changes that commonly coexist with Alzheimer's hallmarks.

These criteria have subsequently been accepted as the standard approach to neuropathologic evaluation of AD and as the gold standard definition of the disease.

Bradley T. Hyman et al. "National Institute on Aging – Alzheimer's Association guidelines on neuropathologic assessment of Alzheimer's disease." *Alzheimer's & Dementia: The Journal of the Alzheimer's Association* 2012;8(1):1 – 13.

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